Name: ______________________

STRAIGHT-LINE GRAPHS, (PARALLEL AND PERPENDICULAR LINES)

Date: ________________

LO: I can identify and find equations of parallel and perpendicular lines.

Parallel and perpendicular

Parallel lines have the **same gradient**. They never meet. **Perpendicular lines** meet at right angles (90°).

Key rule: If one line has gradient m, a perpendicular line has gradient **-1/m** (negative reciprocal). The product of their gradients is -1.

Quick examples

	y = 2x + 1 and y = 2x - 3 are parallel	same gradient
	m&sub1; × m&sub2; = -1 for perpendicular	product rule
	gradient 3 perpendicular gradient $-\frac{1}{3}$	negative reciprocal
	y = 2x + 1 and y = -2x + 3 are perpendicular	2 × (-2) = -4 ≠ -1

Key idea

Parallel: same gradient. Perpendicular: gradients multiply to give -1 (negative reciprocal).

m&sub1; × m&sub2; = -1

m&sub2; = -1/m&sub1;

Perpendicular gradients are negative reciprocals

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – parallel lines

Find the equation of the line parallel to y = 3x + 2 passing through (1, 7).

Parallel means same gradient: m = 3

y = 3x + c; substitute (1, 7): 7 = 3 + c, c = 4

Equation: y = 3x + 4

y = 3x + 4

Parallel lines share the same gradient.

Example 2 – perpendicular gradient

A line has gradient 4. Find the gradient of a perpendicular line.

Perpendicular gradient = -1/m

$-\frac{1}{4}$

The perpendicular gradient is $-\frac{1}{4}$

Gradient = $-\frac{1}{4}$

Flip the fraction and change the sign.

Example 3 – perpendicular equation

Find the equation perpendicular to y = 2x - 5 through (4, 3).

Original gradient = 2; perpendicular = $-\frac{1}{2}$

y = $-\frac{1}{2}$x + c; 3 = $-\frac{1}{2}$(4) + c; 3 = -2 + c; c = 5

y = $-\frac{1}{2}$x + 5

y = $-\frac{1}{2}$x + 5

Find the perpendicular gradient, then use the point to find c.

Example 4 – checking perpendicular

Are y = 3x + 1 and y = $-\frac{1}{3}$x + 4 perpendicular?

m&sub1; = 3, m&sub2; = $-\frac{1}{3}$

Product = 3 × ($-\frac{1}{3}$) = -1

Yes, they are perpendicular

Yes (product of gradients = -1)

If m&sub1; × m&sub2; = -1, the lines are perpendicular.

YOUR TURN – PRACTICE (deliberate practice)

1. Identifying parallel lines
State which pairs are parallel.

a) $y = 5x + 2$ and $y = 5x - 1$

b) $y = 3x + 4$ and $y = -3x + 4$

c) $y = \frac{1}{2}x + 1$ and $y = \frac{1}{2}x - 6$

d) $y = 2x + 3$ and $2y = 4x + 1$

2. Finding perpendicular gradients
Find the perpendicular gradient.

a) $m = 2$

b) $m = -3$

c) $m = \frac{1}{4}$

d) $m = -\frac{2}{5}$

3. Equations of parallel lines
Find the equation of the parallel line through the given point.

a) Parallel to $y = 4x + 1$, through (2, 11)

b) Parallel to $y = -x + 6$, through (3, 1)

c) Parallel to $y = \frac{1}{2}x - 3$, through (4, 5)

d) Parallel to $y = 3x$, through (1, 8)

4. Equations of perpendicular lines
Find the equation perpendicular to the given line through the given point.

a) Perpendicular to $y = 2x + 1$, through (4, 3)

b) Perpendicular to $y = -3x + 2$, through (6, 4)

c) Perpendicular to $y = \frac{1}{5}x + 3$, through (5, -2)

d) Perpendicular to $y = x - 4$, through (2, 6)

5. Mixed problems

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Parallel lines have the same gradient ☐	**b) Perpendicular gradients add to -1** ☐	**c) If m&sub1; × m&sub2; = -1, the lines are perpendicular** ☐	**d) The perpendicular gradient of 2 is -2** ☐	**e) Parallel lines never intersect** ☐
Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________
Reason: _________________	Reason: _________________	Reason: _________________	Reason: _________________	Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Triangle ABC has vertices **A(1, 2)**, **B(5, 4)**, **C(3, 8)**. Show that AB is perpendicular to BC. Find the equations of lines AB and BC.

Expression: ____________________________

Simplified: ____________________________

b) The perpendicular bisector of a line segment passes through the midpoint at right angles. Find the equation of the perpendicular bisector of the segment joining **(2, 3)** and **(6, 7)**.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!